

Topic: 8086 in Minimum Mode

Q. Discuss the configuration and functioning of the 8086 microprocessor in Minimum Mode. Detail the generation and purpose of key control signals.

> **Definition to Remember**

8086 Microprocessor in Minimum Mode

- Acts as the sole controller of the system bus.

2. Key Minimum Mode Control Signals

The 8086 provides dedicated pins for system control:

3. Generation of Specific Control Signals

Because the 8086 outputs generic ``M/IO``, ``RD``, and ``WR`` signals, they must be combined using external logic gates (like a 3-to-8 decoder) to generate specific signals:

4. Basic Operation Cycle

1. CPU outputs the address and pulses **ALE** high. External latches hold the address.

> **Must-Write Points (for 10 marks)**

> **Quick Recall**

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